Unit 3 - Session 1

SLO 2: Explain the mathematical foundation of the Hough Transform.

UG Level | Duration: 10 Minutes per Activity

Question:

Question 1

Given an image point $P(4,3)$, what is the equation used in the Hough Transform to represent a straight line?
Ans- ρ = x cos θ + y sin θ

Question 2

For the point $P(4,3)$, compute the value of ρusing the Hough Transform equation for the following angles:

$$\theta = 0°, 45°, 90°$$

Ans- Substitute the given point (x, y) into ρ = x cos θ + y sin θ for each given angle θ.

Question 3

What does the set of computed (ρ, θ)values for a single image point represent in Hough space?
Ans- The set of computed ρ values for different θ represents a sinusoidal curve in Hough space for the point P(4,3).

Question 4

A second point $Q(8,6)$is given.
If points $P$and Qlie on the same straight line, what should happen mathematically in Hough space?

Ans- If P(4,3) and Q(8,6) lie on the same straight line, their curves in Hough space intersect at a common (ρ, θ) point.

Question 5

How does the Hough Transform use an accumulator (voting table) to detect lines?

Ans- The Hough Transform uses an accumulator (voting table) where each image point votes for possible (ρ, θ) values. A high number of votes at a location indicates a possible line.

Question 6

What does the highest vote (peak) in the accumulator correspond to?

Ans- The highest vote (peak) in the accumulator corresponds to the most likely straight line in the image.

Question 7

What is the overall mathematical interpretation of the Hough Transform in line detection?

Ans- The Hough Transform mathematically converts points in image space into line parameters (ρ, θ) in Hough space. Lines are detected by finding peaks where many points vote for the same parameters.